

Supplementary Material: Efficient Transformer Architectures for Electrodermal Activity-based Arousal Classification

Roberto Sánchez-Reolid, Francisco Javier Celdrán, Daniel Sánchez-Reolid, and
Antonio Fernández-Caballero

No Institute Given

Table S1: Channel Ablation Analysis

Table 1 presents the complete channel ablation results under the Leave-One-Subject-Out protocol, evaluating three input configurations for all eight architectures. Values represent mean F1-score across 147 folds. Percentage values in parentheses indicate the relative improvement over the SCR-only baseline.

Table 1. Channel ablation results: F1-score under LOSO for three input configurations.

Model	SCR only	SCR + Δ SCR	SCR + Δ SCR + Δ^2 SCR
DLinear	0.778	0.792 (+1.8%)	0.800 (+2.8%)
ModernTCN	0.805	0.819 (+1.7%)	0.827 (+2.7%)
FEDformer	0.816	0.829 (+1.6%)	0.836 (+2.5%)
Informer	0.827	0.839 (+1.5%)	0.845 (+2.2%)
Autoformer	0.830	0.842 (+1.4%)	0.848 (+2.2%)
TimesNet	0.836	0.847 (+1.3%)	0.853 (+2.0%)
Mamba	0.842	0.853 (+1.3%)	0.858 (+1.9%)
PatchTST	0.848	0.858 (+1.2%)	0.863 (+1.8%)

Two consistent patterns emerge. First, the inclusion of derivative channels yields incremental but systematic F1 improvements across all architectures. The first derivative (Δ SCR) contributes the largest single gain (+1.0–1.4 pp), while the second derivative (Δ^2 SCR) provides an additional smaller improvement (+0.5–0.8 pp). This confirms that velocity and acceleration of sympathetic activation encode information complementary to the phasic amplitude alone.

Second, the relative contribution of derivative information is inversely related to the architecture’s temporal modelling capacity. Architectures with limited temporal modelling (DLinear, ModernTCN) benefit most from explicit derivative channels (+2.7–2.8% total improvement), while architectures with adaptive global receptive fields (Mamba, PatchTST) show smaller relative gains (+1.8–1.9%). This suggests that attention and state-space mechanisms partially learn

temporal derivative information implicitly from the raw SCR signal, whereas fixed-receptive-field architectures rely more heavily on explicit derivative encoding.

Table S2: Pairwise Statistical Significance

Table 2 presents the complete matrix of pairwise Wilcoxon signed-rank test p -values on per-fold F1-scores (147 folds). Bold values indicate statistical significance at the conventional $\alpha = 0.05$ level (raw, uncorrected). The Bonferroni-Holm corrected threshold for the 28 comparisons is $p_{\text{BH}} < 0.0018$.

Table 2. Pairwise Wilcoxon signed-rank test p -values on per-fold F1-scores.

	DLinear	ModernTCN	FEDformer	Informer	Autoformer	TimesNet	Mamba
PatchTST	$< 10^{-4}$	0.0003	0.0004	0.0012	0.0015	0.0028	0.048
Mamba	0.0002	0.0005	0.0008	0.021	0.032	0.085	
TimesNet	0.0003	0.0006	0.004	0.058	0.072		
Autoformer	0.0004	0.005	0.038	0.065			
Informer	0.0005	0.008	0.045				
FEDformer	0.0008	0.032					
ModernTCN	0.002						

Under the conventional $\alpha = 0.05$ threshold, 24 of 28 comparisons are statistically significant. The most notable result is the PatchTST–Mamba comparison ($p = 0.048$), which is significant at the conventional level but does not survive the stringent Bonferroni-Holm correction ($p_{\text{BH}} < 0.0018$). Under the corrected threshold, 13 of 28 comparisons remain significant. Comparisons between architectures within the same paradigm family (Informer vs. Autoformer, $p = 0.065$; Autoformer vs. TimesNet, $p = 0.072$; Mamba vs. TimesNet, $p = 0.085$) do not survive either threshold, consistent with their shared computational mechanisms and similar performance levels.

Table S3: Per-Class Performance

Table 3 reports per-class F1-scores for the calm and stress conditions, enabling assessment of class-specific performance. The asymmetry in arousal classification difficulty is reflected in the F1 differences between conditions.

A consistent pattern emerges: all architectures achieve slightly higher F1 on the stress condition (mean $+0.8$ pp across architectures), suggesting that sympathetic activation patterns during stress states are more discriminative than those during calm states. This asymmetry is consistent with the physiological role of EDA as a sympathetic activation marker—calm states exhibit lower and more variable SCR amplitudes, while stress states produce more pronounced

Table 3. Per-class F1-scores under LOSO (mean across 147 folds).

Model	F1 (Calm)	F1 (Stress)
DLinear	0.795	0.805
ModernTCN	0.822	0.832
FEDformer	0.831	0.841
Informer	0.841	0.849
Autoformer	0.844	0.852
TimesNet	0.849	0.857
Mamba	0.855	0.861
PatchTST	0.860	0.866

and stereotyped phasic responses [1]. The stress-class advantage is most pronounced in simpler architectures (DLinear: +1.0 pp) and narrows in those with stronger temporal modelling (PatchTST: +0.6 pp, Mamba: +0.6 pp), indicating that global dependency modelling partially compensates for the reduced discriminability of calm-state signals.

Table S4: Training Time and Convergence

Table 4 reports training time per epoch, epochs to convergence, and total training wall-clock time for each architecture on the NVIDIA RTX 4080 (16 GB GDDR6X, Ada Lovelace). All measurements use the optimal hyperparameter configuration, batch size 32, and the full 147-subject training set. Early stopping with patience 15 was applied; convergence is defined as the epoch at which validation loss stopped improving.

Table 4. Training time and convergence metrics on RTX 4080. Epochs reported are means across 147 LOSO folds.

Model	t_{train} (s/epoch)	Epochs (mean)	Total training (h)
DLinear	1.2	32	0.6
ModernTCN	8.5	45	6.4
Mamba	12.3	58	11.9
FEDformer	18.7	62	19.3
TimesNet	22.4	55	20.5
Informer	28.1	68	31.8
Autoformer	35.6	72	42.7
PatchTST	42.8	85	60.6

Training time follows the same hierarchy as inference time, with DLinear converging approximately 100 $\times$ faster than PatchTST in total wall-clock time (0.6 h vs. 60.6 h). The increasing number of epochs required for convergence as

architectural complexity grows reflects the additional optimisation difficulty introduced by attention mechanisms, state space models, and deeper architectures. For practitioners, this translates to substantially different resource requirements for hyperparameter tuning: the 64-configuration grid search described in the main text requires approximately 3 GPU-hours for DLinear versus over 100 GPU-hours for PatchTST on the RTX 4080. These estimates account for the fact that not all configurations run to full convergence—configurations with poor validation performance are terminated early by the patience-based stopping criterion, reducing the effective per-configuration cost to approximately 40–60% of the full-convergence time.

References

1. Boucsein, W.: *Electrodermal Activity*. Springer, 2nd edn. (2012). <https://doi.org/10.1007/978-1-4614-1126-0>

Further Reading

The following references, while not directly cited in the main text, provide relevant context for the methods and datasets discussed in this work.

EDA and Affective Computing Foundations

- **Boucsein (2012)** — *Electrodermal Activity* (2nd ed., Springer). The definitive textbook on EDA measurement, signal characteristics, and physiological interpretation.
- **Picard (1997)** — *Affective Computing* (MIT Press). Foundational work establishing the field of affective computing and the role of physiological signals in emotion recognition.
- **Meijer & Verschuere (2023)** — The physiological basis of electrodermal activity. *Biological Psychology*, 178, 108530. Reviews the neurophysiological mechanisms underlying EDA.

Efficient and Lightweight Architectures

- **Tay et al. (2022)** — Efficient transformers: A survey. *ACM Computing Surveys*, 55(6), 1–28. Comprehensive survey of efficient transformer architectures.
- **Vaswani et al. (2017)** — Attention is all you need. *NeurIPS*, 30. The original Transformer architecture that motivated the efficient variants evaluated in this work.
- **Gopi (2025)** — Lightweight transformer models for biomedical signal processing: Trends, challenges, and future directions. Survey of lightweight architectures for biosignals.
- **Saravana et al. (2026)** — Transformers for multivariate time series forecasting: Comprehensive analysis. *IEEE Access*. Systematic analysis of transformer variants including PatchTST, Autoformer, and FEDformer.

Time-Series Classification and Biosignal Processing

- **Ismail Fawaz et al. (2020)** — InceptionTime: Finding AlexNet for time series classification. *Data Mining and Knowledge Discovery*, 34(6), 1936–1962. The InceptionTime architecture used as a baseline in our prior work.
- **Luo et al. (2024)** — Temporal convolutional networks for physiological time-series: A review. *IEEE Access*. Reviews TCN applications to biosignals.
- **Nandipati et al. (2024)** — CNN-based analysis of skin conductance responses. *Computers in Biology and Medicine*. Convolutional approaches specifically for EDA signals.
- **Wang et al. (2024)** — Self-supervised learning with transformers for physiological signals. *Nature Biomedical Engineering*. Self-supervised pretraining strategies relevant to future work directions.

Benchmarking and Comparative Studies

- **Pativada (2024)** — Exploring the effectiveness of transformers for long-term time series forecasting. Independent benchmark comparing Informer, Autoformer, FEDformer, and PatchTST on forecasting tasks.
- **Portal et al. (2025)** — Performance evaluation of deep learning architectures for physiological signal classification. *Biomedical Signal Processing and Control*. Recent benchmark of architectures for biosignal classification.
- **Yaseen & Şahin (2026)** — Masked signal modeling with self-supervised transformers for stress and affect detection using wearable biosignals. *Biomedical Signal Processing and Control*. Self-supervised approach to stress detection from physiological signals.